

Moderately Exponential Time Approximation Algorithms for the Maximum Bounded-degree-1 Set Problem*

Maw-Shang Chang and Ling-Ju Hung[†]

Department of Computer Science and Information Engineering
HungKuang University, Taichung 43302, Taiwan
{mschang, ljhung}@sunrise.hk.edu.tw

Abstract

Given a graph $G = (V, E)$, a bounded-degree-1 set S is a vertex subset of G such that the maximum degree in $G[S]$ is at most one. The MAXIMUM BOUNDED-DEGREE-1 SET (MAX 1-BDS) problem is to find a bounded-degree-1 set S of maximum cardinality in G . It is an NP-hard problem. In this paper, we show that unless $P = NP$ the MAX 1-BDS problem can not be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time for all $\epsilon > 0$. We design a series of moderately exponential time r -approximation algorithms to solve the MAX 1-BDS where $0 < r < 1$ is a constant. The dual problem of MAX 1-BDS problem, MINIMUM BOUNDED-DEGREE-1 DELETION SET (MIN 1-BDD) problem, is a generalized MINIMUM VERTEX COVER problem. It was shown in the literature that unless $P = NP$, there is no c -approximation algorithm for MINIMUM VERTEX COVER problem for any $c < 1.36$. We give a $\frac{4}{3}$ -approximation algorithm for the MIN 1-BDD problem running in time $O^*(1.4214^n)$.

1 Introduction

Approximation algorithms are practical because they provide solutions with good qualities. Many NP-hard problems have polynomial time α -approximation algorithms. For example, the MINIMUM VERTEX COVER problem has a trivial 2-approximation algorithm. However, there are NP-hard problems that can not be approximated within α unless $P = NP$. The MINIMUM VERTEX COVER problem is APX-complete [17].

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Moreover, it was shown NP-hard to approximate to within any factor smaller than 1.36 [9]. We can not expect that MINIMUM VERTEX COVER problem admits a polynomial time approximation scheme (PTAS) or has a α -approximation algorithm for any $\alpha < 1.36$ running in polynomial time. The MAXIMUM INDEPENDENT SET problem is the dual problem of the MINIMUM VERTEX COVER problem. It is shown that the MAXIMUM INDEPENDENT SET cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time for any $\epsilon > 0$ if $P \neq NP$ [24]. In this circumstance, exponential-time algorithms are another choice to design approximation algorithms. Bourgeois *et al.* [6] gave a series of moderately exponential time approximation algorithms for the MAXIMUM INDEPENDENT SET problem and the MINIMUM VERTEX COVER problem that break the barrier of the approximation factor.

A bounded-degree- d set S in an undirected graph $G = (V, E)$ is a vertex subset such that the maximum degree of $G[S]$ is at most d . The MAXIMUM BOUNDED-DEGREE- d SET problem is to find a bounded-degree- d set S of maximum size in the input graph $G = (V, E)$. A k -plex S in a graph $G = (V, E)$ is a vertex subset such that for each $v \in S$, $\deg_{G[S]}(v) \geq |S| - k$. We use P_ℓ to denote a simple path having ℓ vertices. Let $N^i(v)$ denote the set of vertices at distance i from v .

In the following, we list the related problems of the MAXIMUM-BOUNDED-DEGREE- d SET problem.

MAXIMUM BOUNDED-DEGREE- d SET (MAX d -BDS)

Input: A graph $G = (V, E)$.

Output: A vertex set $S \subseteq V$ of maximum cardinality such that S is bounded-degree- d set.

MAXIMUM k -PLEX (MAX k -PLEX)

Input: A graph $G = (V, E)$.

Output: A vertex set $S \subseteq V$ of maximum cardinality such that S is a k -plex.

nality such that S is a k -plex.

MINIMUM BOUNDED-DEGREE- d DELETION SET (MIN d -BDD)

Input: A graph $G = (V, E)$.

Output: A vertex set $D \subseteq V$ of minimum cardinality such that $V \setminus D$ is bounded-degree- d set.

MINIMUM VERTEX COVER P_ℓ (MIN P_ℓ -VC)

Input: A graph $G = (V, E)$.

Output: A vertex set $C \subseteq V$ of minimum cardinality such that $G[V \setminus C]$ has no P_ℓ .

Note that a vertex subset S is a k -plex in G if and only if it is a bounded-degree- $(k-1)$ set in $\bar{G}$.

The MAXIMUM BOUNDED-DEGREE- d SET (MAX d -BDS) problem is NP-complete because the MAXIMUM $(d+1)$ -PLEX (MAX $(d+1)$ -PLEX) problem is NP-complete [2]. The MINIMUM BOUNDED-DEGREE- d DELETION SET (MIN d -BDD) problem is the dual problem of the MAX d -BDS problem. A series of fixed-parameter algorithms were developed for solving the MIN- d -BDD problem by taking the size of the solution as the input parameter [16, 12, 14, 10]. Betzler *et al.* [4] showed that when parameterized by treewidth the MIN d -BDD problem is W[1]-hard. They also showed that the MIN d -BDD problem is fixed-parameter tractable when parameterized by the combined parameter treewidth and number of vertices to delete and when parameterized by the feedback edge set number [4].

The model k -plex is defined as a degree based relaxed model for finding cohesive subgroups in social networks [19]. The MAX k -PLEX problem can be formulated in 0/1 integer programming [1, 2]. It is shown W[1]-hard with respect to the parameter s being the size of k -plexes [12]. Some branch-and-bound algorithms were given for solving the MAX k -PLEX problem based on different upper bounds and lower bounds found by heuristics algorithms [20, 13]. Some graph editing problems are studied on finding a disjoint union of k -plexes [11, 5]. Balasundaram *et al.* [3] referred the Max d -bds problem as the problem of finding maximum-cardinality co- $(d+1)$ -plexes. Wu and Pei [23] gave an algorithm to enumerate all maximal k -plexes.

The MINIMUM VERTEX COVER P_ℓ (MIN P_ℓ -VC) problem is NP-complete for $\ell \geq 2$ [21]. It is a generalized MINIMUM VERTEX COVER problem. The MINIMUM P_2 -VC problem is equivalent to the MINIMUM VERTEX COVER problem. Tu and Zhou [21] gave a 2-approximation algorithm for the MIN P_3 -VC problem using linear programming approaches. Later, they gave a greedy algo-

rithm for the same problem and showed that the algorithm is a 2-approximation algorithm [22].

In this paper, we show that unless $P = NP$ the MAX 1-BDS problem and the MAX 2-PLEX problem can not be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time for all $\epsilon > 0$. We design a series of moderately exponential time r -approximation algorithms to solve the MAX 1-BDS where $0 < r < 1$ is a constant. The dual problem of MAX 1-BDS problem, the MIN 1-BDD problem, is a generalized MINIMUM VERTEX COVER problem. Notice that to approximate the solution of the MINIMUM VERTEX COVER problem to within any factor smaller than 1.36 is NP-hard [9]. We give a $\frac{4}{3}$ -approximation algorithm for the MIN 1-BDD problem running in time $O^*(1.4214^n)$.

This paper is organized as follows. In Section 2, we describe our inapproximability results of the MAX 1-BDS problem and the MAX 2-PLEX problem. In Section 3, we apply four different strategies to design moderately exponential time approximation algorithms for the MAX 1-BDS problem and MIN 1-BDD problem. Finally, we give concluding remarks in Section 4.

We end this section with some notation. A modified O -notation, O^* , is used here to bound the running time of exponential time algorithms asymptotically. For functions f and g , $f(n) = O^*(g(n))$ if $f(n) = O(g(n) \cdot \text{poly}(n))$ where $\text{poly}(n)$ is a polynomial. Given a graph $G = (V, E)$, we use n to denote the number of vertices in G . Let $N(v)$ be the set of vertices adjacent to v and $N[v] = N(v) \cup \{v\}$. For a vertex subset X , $N[X] = \bigcup_{v \in X} N[v]$ and $N(X) = N[X] \setminus X$. Let $\deg(v) = |N(v)|$ to denote the degree of v . A vertex v is called a *degree-0 vertex* if $\deg(v) = 0$ and is called a *degree-one vertex* if $\deg(v) = 1$. We use $N^i(v)$ to denote the set of vertices at distance i from v . For a vertex v , we use $G - v$ to denote $G[V \setminus \{v\}]$. For a vertex subset X , we use $G - X$ to denote $G[V \setminus X]$. Let $\Delta(G)$ be the maximum degree of G .

2 Inapproximability results

In this section, we show the inapproximability of the MAX 1-BDS problem and the MAX 2-PLEX problem.

Theorem 1. *If $P \neq NP$, for all $\epsilon > 0$ the MAX 1-BDS problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time.*

Proof. Suppose that the MAX 1-BDS problem can

be approximated to a ratio r in polynomial time, $r > n^{\epsilon-1}$ for some $\epsilon > 0$. Let S be the solution found by the approximation algorithm and S^* be an optimal solution. Then

$$\frac{|S|}{|S^*|} \geq r.$$

Let I^* be a maximum independent set in the input graph. Since I^* is also a bounded-degree-1 set, $|I^*| \leq |S^*|$. We see that $G[S^*]$ consists of degree-1 vertices and degree-0 vertices. One can obtain an independent set from S^* by removing half degree-1 vertices in $G[S^*]$. Similarly, we can obtain an independent set I from S by removing half degree-1 vertices in $G[S]$ from S . It is easy to see that

$$|I| \geq \frac{|S|}{2} \geq \frac{r \cdot |S^*|}{2} \geq \frac{r \cdot |I^*|}{2}.$$

This implies

$$\frac{|I|}{|I^*|} \geq \frac{r}{2} > \frac{n^{\epsilon-1}}{2} = n^{\epsilon'-1}$$

where $\epsilon' = \epsilon - \frac{1}{\lg n} > 0$ for $n > 2^{\frac{1}{\epsilon}}$.

Thus, the MAXIMUM INDEPENDENT SET problem can be approximated to a ratio greater than $n^{\epsilon'-1}$ for some $\epsilon' > 0$. This contradicts to the fact that the MAXIMUM INDEPENDENT SET problem cannot be approximated to a ratio greater than $n^{\epsilon'-1}$ in polynomial time for any $\epsilon' > 0$ if $P \neq NP$ [24]. Therefore, if $P \neq NP$, for any $\epsilon > 0$ the MAX 1-BDS problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time. $\square$

Corollary 1. *If $P \neq NP$, for all $\epsilon > 0$ the MAX 2-PLEX problem cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time.*

Proof. Suppose the MAX 2-PLEX problem can be approximated to a ratio r in polynomial time where $r > n^{\epsilon-1}$ for some $\epsilon > 0$. Given a graph G , let S be the solution found by the approximation algorithm in G and S^* be an optimal solution in G . Then

$$\frac{|S|}{|S^*|} \geq r.$$

Notice that S^* is a bounded-degree-1 set of maximum size in $\bar{G}$ and S is a bounded-degree-1 set in $\bar{G}$. This implies that the MAX 1-BDS problem can be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time, a contradiction to the fact that the MAX 1-BDS cannot be approximated to a ratio greater than $n^{\epsilon-1}$ in polynomial time. This completes the proof. $\square$

3 Moderately exponential time approximation algorithms

In this section, we give some moderately exponential time approximation algorithms for the MAX 1-BDS problem and its related problems listed in Section 1.

Notice that solving the MAX 1-BDS problem in G is equivalent to solving the MAX 2-PLEX problem in $\bar{G}$. It is straightforward that if the MAX 1-BDS problem can be solved in time $O^*(\gamma^n)$ time, then the MAX 2-PLEX problem can be solved in time $O^*(\gamma^n)$. In the following theorem, we give a further result.

Lemma 1. *If the MAX 1-BDS problem can be solved in time $O^*(\gamma^n)$ time, then the MAX 2-PLEX problem can be solved in time $O^*(\gamma^{\Delta(G)})$.*

Proof. Let $G = (V, E)$ be an input graph of the MAX 2-PLEX problem and S be a 2-plex of maximum size in G . By definition, for any $v \in S$, there exists at most one vertex $x \in S$ that is not adjacent to v but is adjacent to some vertex in $N(v)$. Thus, for each $v \in V$ and each $x \in N^2(v)$, we construct $G_{vx} = G[N[v] \cup \{x\}]$ and for each G_{vx} we run the algorithm for solving the MAX 1-BDS problem on $\bar{G}_{vx}$. Finally, return a bounded-degree-1 set S of maximum cardinality among all $\bar{G}_{vx}$. It is easy to see that S is a maximum 2-plex in G . Since there are at most $\Delta(G) + 2$ vertices in G_{vx} , it takes $O^*(\gamma^{\Delta(G)})$ time to find a maximum bounded-degree-1 set in each $\bar{G}_{vx}$. Note that for each v and for each $x \in N^2(v)$ we construct a graph G_{vx} . In total, we construct at most $\binom{n}{2}$ graphs. Thus, the running time of the algorithm for the MAX 2-PLEX problem is

$$O^* \left(\binom{n}{2} \cdot \gamma^{\Delta(G)} \right) = O^*(\gamma^{\Delta(G)}).$$

This completes the proof. $\square$

Remark 1. *Chang et al. [8] gave an exact algorithm for the MAX 1-BDS problem running in time $O^*(1.4834^n)$. By applying this algorithm, the MAX 2-PLEX problem can be solved in time $O^*(1.4834^{\Delta(G)})$.*

According to Lemma 1 and the observation that S is a bounded-degree-1 set in G iff it is a 2-plex in $\bar{G}$, the following lemma holds directly.

Lemma 2. *If the MAX 1-BDS problem is r -approximable in time $O^*(\gamma^n)$, then MAX 2-PLEX problem is r -approximable in time $O^*(\gamma^{\Delta(G)})$.*

3.1 Apply an exact algorithm

In this section, we show that one can apply moderately exponential time algorithms for the MAXIMUM INDEPENDENT SET problem to design an approximation algorithm for the MAX 1-BDS problem and the MAX 2-PLEX problem.

It is easy to see that any independent set is also a bounded-degree-1 set. In the following theorem, we show that a maximum independent set is a $1/2$ -approximation solution for the MAX 1-BDS problem.

Theorem 2. *Any exact algorithm for the MAXIMUM INDEPENDENT SET problem is a $1/2$ -approximation algorithm for the MAX 1-BDS problem.*

Proof. Let I^* be the maximum independent set in the input graph G found by an exact algorithm for the MAXIMUM INDEPENDENT SET problem. Suppose that S^* is a bounded degree-1 set of maximum size in G . Notice that I^* is a bounded degree-1 set. Thus $|I^*| \leq |S^*|$. Let $D_1 \subseteq S^*$ and $D_0 \subseteq S^*$ denote the set of degree-1 vertices and degree-0 vertices in $G[S]$, respectively. We have $S^* = D_1 \cup D_0$ and $|S^*| = |D_1| + |D_0|$. Let D be a maximum independent set in $G[D_1]$. It is easy to see that $|D| = |D_1|/2$ and $D \cup D_0$ forms an independent set in G . Thus

$$|I^*| \geq |D| + |D_0| = \frac{|D_1|}{2} + |D_0| \geq \frac{|S^*|}{2}.$$

We have

$$\frac{|I^*|}{|S^*|} \geq \frac{1}{2}.$$

This completes the proof. $\square$

Corollary 2. *If there is an r -approximation algorithm for the MAXIMUM INDEPENDENT SET problem, then this algorithm is an $\frac{r}{2}$ -approximation algorithm for the MAX 1-BDS problem.*

Proof. Let I^* be an independent set of maximum size in $G = (V, E)$ and S^* be a bounded-degree-1 set of maximum size in G . Consider an r -approximation algorithm A solving the MAXIMUM INDEPENDENT SET problem. Let I be the solution found by algorithm A . Note that I is also a bounded-degree-1 set in G . Thus

$$\frac{|I|}{|S^*|} \geq \frac{r \cdot |I^*|}{|S^*|} \geq \frac{r}{2}.$$

This completes the proof. $\square$

Remark 2. Robson [18] gave an $O^*(1.2109^n)$ -time algorithm for the MAXIMUM INDEPENDENT SET problem. According to the above theorem, the MAX 1-BDS problem has a $1/2$ -approximation algorithm running in time $O^*(1.2109^n)$.

Remark 3. Bourgeois et al. [6] gave a series of moderately exponential time r -approximation algorithms for the MAXIMUM INDEPENDENT SET problem, all of them are $r/2$ -approximation algorithms for the MAX 1-BDS problem. For example, they gave $1/2$ -approximation algorithm for the MAXIMUM INDEPENDENT SET problem running in time $O^*(1.086^n)$. According to the above corollary, the MAX 1-BDS problem has a $1/4$ -approximation algorithm running in time $O^*(1.086^n)$. By Lemma 2 the MAX 2-PLEX problem has a $1/4$ -approximation algorithm running in time $O^*(1.086^{\Delta(G)})$ where G is an input instance of the MAX 2-PLEX problem and $\Delta(G)$ is the maximum degree of G .

3.2 Divide and approximate

Bourgeois et al. [6] showed that if there exists an exact algorithm for the MAXIMUM INDEPENDENT SET problem running in time $O^*(\gamma^n)$, there exists p/q -approximation algorithm for the MAXIMUM INDEPENDENT SET problem that runs in $O^*(\gamma^{\frac{p}{q}n})$ for any positive integers $p \leq q$. Moreover, it also holds for any maximization problem satisfying some *hereditary* property [6]. In the following theorem, we show that one can design approximation algorithms for the MAX 1-BDS by using the same approach given in [6].

Theorem 3. *If the MAX 1-BDS problem can be solved in time $O^*(\gamma^n)$ time, then there exists an p/q -approximation algorithm that runs in time $O^*(\gamma^{\frac{p}{q}n})$ for the MAX 1-BDS problem for any two positive integers $p \leq q$.*

Proof. Consider a graph $G = (V, E)$ of n vertices. Let 1-BDS be the algorithm running in time $O^*(\gamma^n)$ for the MAX 1-BDS problem. We have the following algorithm.

1. Partition V into $(V_1, V_2, \dots, V_q)$ where $|V_i| \leq \lceil n/q \rceil$.
2. Construct $G_1, G_2, \dots, G_q$ where $G_i = G[\bigcup_{j=i}^{i+p-1} V_{(j \bmod q)}]$.
3. Run 1-BDS on G_i , $i = 1, \dots, q$.
4. Output the best solution among all solutions found by algorithm 1-BDS on G_i , $i = 1, \dots, q$.

Let S be the solution found by the above algorithm and S^* be an optimal solution of G . For each $i = 1, \dots, q$,

$$|S^* \cap (\bigcup_{j=i}^{i+p-1} V_{(j \bmod q)})| \leq |S|.$$

This implies

$$\sum_{i=1}^q |S^* \cap (\bigcup_{j=i}^{i+p-1} V_{(j \bmod q)})| \leq q \cdot |S|.$$

Thus

$$p \cdot |S^*| \leq q \cdot |S|$$

and

$$\frac{|S|}{|S^*|} \geq \frac{p}{q}.$$

Since each G_i has at most $\lceil pn/q \rceil$ vertices, the running time of the approximation algorithm is $O^*(\gamma^{\frac{p}{q} \cdot n})$. This completes the proof. $\square$

Corollary 3. *If the MAX 1-BDS problem can be solved in time $O^*(\gamma^n)$ time, then there exists a p/q -approximation algorithm that runs in time $O^*(\gamma^{\frac{p}{q} \cdot \Delta(G)})$ for the MAX 2-PLEX problem.*

Proof. By Theorem 3, if there exists an exact algorithm for the MAX 1-BDS problem running in time $O^*(\gamma^n)$, then there exists a p/q -approximation algorithm for the MAX 1-BDS problem running in time $O^*(\gamma^{\frac{p}{q} \cdot n})$ for any two positive integers $p \leq q$. By Lemma 2, we see that there exists a p/q -approximation algorithm for the MAX 2-PLEX problem running in time $O^*(\gamma^{\frac{p}{q} \cdot \Delta(G)})$. This completes the proof. $\square$

3.3 Apply a kernelization algorithm

One can apply a kernelization algorithm to obtain a size- $2k$ kernel for the MINIMUM VERTEX COVER problem where k is the size of vertex cover [15]. This implies we can always assume that the size of minimum vertex cover is at least $n/2$. Suppose that we have an instance $G = (V, E)$ of the MINIMUM VERTEX COVER problem satisfying the condition that the minimum vertex cover in G is at least $n/2$. Bourgeois *et al.* [6] showed that if there exists a ρ -approximation algorithm for the MAXIMUM INDEPENDENT SET problem, then the MINIMUM VERTEX COVER problem can be approximated within ratio $2 - \rho$. In this section, we show that a similar relation holds between the MAX 1-BDS problem and the MIN 1-BDD problem.

The MIN 1-BDD problem is the dual problem of the MAX 1-BDS problem. In the following lemma, we show that it is equivalent to the MIN P_3 -VC problem.

Lemma 3. *The MIN 1-BDD problem and the MIN P_3 -VC problem are equivalent.*

Proof. Let D be an optimal solution of the MIN 1-BDD problem in the input graph $G = (V, E)$. Let C be an optimal solution of the MIN P_3 -VC problem. Since $G[V \setminus C]$ has no P_3 , it consists of P_1 and P_2 , *i.e.*, degree-0 vertices and degree-1 vertices. Thus C is a feasible solution of the MIN 1-BDD problem. We see that $|C| \geq |D|$. Since $G[V \setminus D]$ consists of degree-1 vertices and degree-0 vertices, it has no P_3 . We see that D is a feasible solution of the MIN P_3 -VC problem. Thus, $|D| \geq |C|$. This shows that $|C| = |D|$ and the two problems are equivalent. $\square$

Theorem 4. *If MAX 1-BDS problem is approximable to an approximation ratio $0 < \rho < 1$ in time $O^*(\gamma^n)$, then MIN 1-BDD (MIN P_3 -VC) is approximable within ratio $15 - 14\rho$ in time $O^*(\gamma^n)$.*

Proof. Fellows *et al.* [10] showed that the MIN 1-BDD problem admits a problem kernel of size at most $15k$ where k is the number of vertices to be removed for obtaining a bounded-degree-1 set and the kernel can be computed in polynomial time. Let D^* be an optimal solution of the MIN 1-BDD problem and $S^* = V \setminus D^*$ be an optimal solution of the MAX 1-BDS problem. We may assume that $n \leq 15 \cdot |D^*|$ and $|S^*| \leq 14n/15$.

Let S be the solution computed by the ρ -approximation algorithm for the MAX 1-BDS problem, $|S| \geq \rho|S^*|$. Let $D = V \setminus S$. Then the approximation ratio is

$$\frac{|D|}{|D^*|} = \frac{n - |S|}{n - |S^*|} \leq \frac{n - \rho|S^*|}{n - |S^*|} = 1 + \frac{1 - \rho}{\frac{n}{|S^*|} - 1}$$

$$\text{Since } \frac{n}{|S^*|} \geq \frac{15}{14},$$

$$\frac{|D|}{|D^*|} \leq 1 + \frac{1 - \rho}{\frac{n}{|S^*|} - 1} \leq 1 + \frac{1 - \rho}{\frac{15}{14} - 1} = 15 - 14\rho.$$

This completes the proof. $\square$

Remark 4. *Suppose that $15 - 14\rho = \frac{4}{3}$. We see that $\rho = \frac{41}{42}$. Note that the MAX 1-BDS problem can be solved optimally in $O^*(1.4834^n)$ [8]. By Theorem 3, there is a $\frac{41}{42}$ -approximation algorithm for the MAX 1-BDS problem running in time $O^*(1.4834^{\frac{41}{42} \cdot n}) = O^*(1.4696^n)$. Thus, the MIN 1-BDD problem has a $\frac{4}{3}$ -approximation algorithm running in time $O^*(1.4696^n)$.*

3.4 Prune the search tree approximately

The strategy *prune the search tree approximately* [6] is used to design approximation algorithms for problems having search tree-based algorithms (also called branching algorithms).

Given a graph G , let D^* be an optimal solution of the MIN 1-BDD problem and $\phi(G) = |D^*|$. Consider the following simple branching algorithm for the MIN 1-BDD problem.

1. If $\Delta(G) \leq 2$, then the MIN 1-BDD problem can be solved in polynomial time [7, 8].
2. If $\Delta(G) \geq 3$ and v is a vertex in G of maximum degree, the algorithm branches into the following cases.
 - (a) Let $D^* = D^* \cup \{v\}$ and solve the problem recursively on $G - v$, i.e., $\phi(G) = \phi(G - v) + 1$.
 - (b) Let $D^* = D^* \cup N(v)$ and solve the problem recursively on $G - N[v]$, i.e., $\phi(G) = \phi(G - N[v]) + |N(v)|$.
 - (c) For all $x \in N(v)$, let $D^* = D^* \cup N(\{v, x\})$ and solve the problem recursively on $G - N[\{v, x\}]$, i.e., $\phi(G) = \phi(G - N[\{v, x\}]) + |N(\{v, x\})|$.

Suppose that the degree of v is d . The running time of the above exact algorithm is

$$T(n) = T(n-1) + (d+1) \cdot T(n-d-1).$$

The worst case happens when $d = 3$ and the running time is $O^*(1.7485^n)$.

Theorem 5. *There exists a $\frac{3}{2}$ -approximation algorithm running in time $O^*(1.3803^n)$ for the MIN 1-BDD problem.*

Proof. Based on the above branching algorithm, we design a $\frac{3}{2}$ -approximation algorithm for the MIN 1-BDD problem. Let D denote the set of the approximation solution found by the following algorithm.

1. If $\Delta(G) \leq 2$, then the MIN 1-BDD problem can be solved in polynomial time [7, 8].
2. If $\Delta(G) \geq 3$, assume that v is a vertex in G of maximum degree, the algorithm branches into the following cases.
 - (a) Let $D = D \cup \{v\}$ and solve the problem recursively on $G - v$, i.e., $\phi(G) = \phi(G - v) + 1$.

- (b) Let $D = D \cup N(v)$ and solve the problem recursively on $G - N[v]$, i.e., $\phi(G) = \phi(G - N[v]) + |N(v)|$.

Suppose that the degree of v is d . The running time of the above approximation algorithm is

$$T(n) = T(n-1) + T(n-d-1).$$

The worst case happens when $d = 3$ and the running time is $O^*(1.3803^n)$.

Now we show that the approximation ratio is $\frac{3}{2}$. Note that when $\Delta(G) \leq 2$, the algorithm does not make any error. Suppose that $\deg(v) = d \geq 3$ and Step 2 is applied. If v is not in the optimal solution D^* , both the approximation algorithm and exact algorithm do not select v in the solution set. If v is a degree-0 vertex in $G[V \setminus D^*]$, the approximation algorithm does not make any error. Suppose that v is a degree-1 vertex in $G[V \setminus D^*]$ and (v, x) is an edge in $G[V \setminus D^*]$. We see that $N(v) \setminus \{x\} \subseteq D^*$. The approximation algorithm only makes an error at Step 2(b) when $x \notin D^*$ but the algorithm selects $x \in D$. Thus,

$$\frac{|D|}{|D^*|} \leq \frac{d}{d-1} \leq \frac{3}{2}.$$

This completes the proof. $\square$

Theorem 6. *There exists a $\frac{4}{3}$ -approximation algorithm running in time $O^*(1.4336^n)$ for the MIN 1-BDD problem.*

Proof. We give the following approximation algorithm for the MIN 1-BDD problem. Let D denote the set of the approximation solution found by the following algorithm.

1. If $\Delta(G) \leq 3$, then optimally solve the MIN 1-BDD problem in time $O^*(1.4336^n)$ [7, 8].
2. If $\Delta(G) \geq 4$, assume that v is a vertex in G of maximum degree, the algorithm branches into the following cases.
 - (a) Let $D = D \cup \{v\}$ and solve the problem recursively on $G - v$, i.e., $\phi(G) = \phi(G - v) + 1$.
 - (b) Let $D = D \cup N(v)$ and solve the problem recursively on $G - N[v]$, i.e., $\phi(G) = \phi(G - N[v]) + |N(v)|$.

Suppose that the degree of v is d . The running time of the algorithm spent on Step 2 is

$$T(n) = T(n-1) + T(n-d-1).$$

Table 1: The main results

Problems	Approximation ratio	Running time
MAX 1-BDS	41/42	$O^*(1.4696^n)$
	1/2	$O^*(1.2109^n)$
	1/4	$O^*(1.086^n)$
MAX 2-PLEX	41/42	$O^*(1.4696^{\Delta(G)})$
	1/2	$O^*(1.2109^{\Delta(G)})$
	1/4	$O^*(1.086^{\Delta(G)})$
MIN 1-BDD	3/2	$O^*(1.3803^n)$
	4/3	$O^*(1.4214^n)$

The worst case happens when $d = 4$ and the running time of the algorithm spent on Step 2 $O^*(1.3248^n)$. The total running time of the approximation algorithm is $O^*(1.4336^n)$ since the algorithm ensures that at some stage the maximum degree of all input instances is at most three. Similar to the proof of Theorem 5, the approximation ratio is $\frac{d}{d-1} \leq \frac{4}{3}$. This completes the proof. $\square$

Remark 5. According to Theorem 4, if we have a $\frac{41}{42}$ -approximation algorithm for the MAX 1-BDS we can obtain a $\frac{4}{3}$ -approximation algorithm for the MIN 1-BDD problem. Note that the MAX 1-BDS problem for graphs of maximum degree-three can be solved in $O^*(1.4336^n)$ [7, 8]. By Theorem 3, there exists a $\frac{41}{42}$ -approximation algorithm for the MAX 1-BDS problem running in time $O^*(1.4336^{\frac{41}{42}n}) = O^*(1.4214^n)$ for graphs of maximum degree three. If we apply this approximation algorithm at Step 1, then we obtain a $\frac{4}{3}$ -approximation algorithm for the MIN 1-BDD problem running in time $O^*(1.4214^n)$.

4 Concluding remarks

In this paper, we design a series of moderately exponential time approximation algorithms for the MAX 1-BDS problem and its related problems. In Table 1, we list these results. We see that the running time of these approximation algorithms depends on the efficiency of a corresponding exact algorithm for the MAX 1-BDS problem. There is an exact algorithm running in time $O^*(1.4834^n)$ for the MAX 1-BDS problem [8]. The open problem is whether the MAX 1-BDS problem can be solved in time $O^*(c^n)$, $c < 1.4834$. Fellows *et al.* [10] gave a size $15k$ kernel for the MIN 1-BDD problem where k is the number of vertices to be removed for obtaining a bounded-degree-1 set. When de-

signing an approximation algorithm for the MIN 1-BDD problem, a linear kernel of size ck , $c < 15$, found by kernelization algorithms helps to improve the approximation ratio. For the feature work, it is interesting to see whether there exists a kernelization algorithm for the MIN 1-BDD problem that computes a kernel of size at most ck for some $c < 15$.

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